

EXPERIMENT 14 – Survey Results Analysis

Question

Analyze survey responses using stacked bar chart, radar chart, and table.

Dataset

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

The supplied experiment requests a stacked bar chart, radar chart, table, and Tableau dashboard; the Tableau part is omitted.

R Code

```
# Experiment 14 - Survey Results
```

```
survey <- data.frame(  
  Respondent = c(1, 2, 3),  
  Question1 = c("A", "B", "C"),  
  Question2 = c("B", "A", "A"),  
  Question3 = c("C", "D", "B")  
)
```

```
# Question 1 - Stacked Bar Chart
```

```
q1 <- table(survey$Question1)  
barplot(q1,  
  main = "Distribution of Question 1 Answers",  
  ylab = "Number of Responses",  
  col = "lightblue")
```

```
# Question 2 - Radar-style Chart
```

```

response_count <- c(
  sum(survey$Question1 == "A"),
  sum(survey$Question1 == "B"),
  sum(survey$Question1 == "C"),
  sum(survey$Question2 == "A"),
  sum(survey$Question2 == "B"),
  sum(survey$Question2 == "C"),
  sum(survey$Question3 == "A"),
  sum(survey$Question3 == "B"),
  sum(survey$Question3 == "C"),
  sum(survey$Question3 == "D")
)

angles <- seq(0, 2*pi, length.out = 11)
values <- c(response_count, response_count[1])
plot(angles, values,
     type = "o",
     xaxt = "n",
     main = "Overall Survey Response Pattern",
     xlab = "",
     ylab = "Response Count")

```

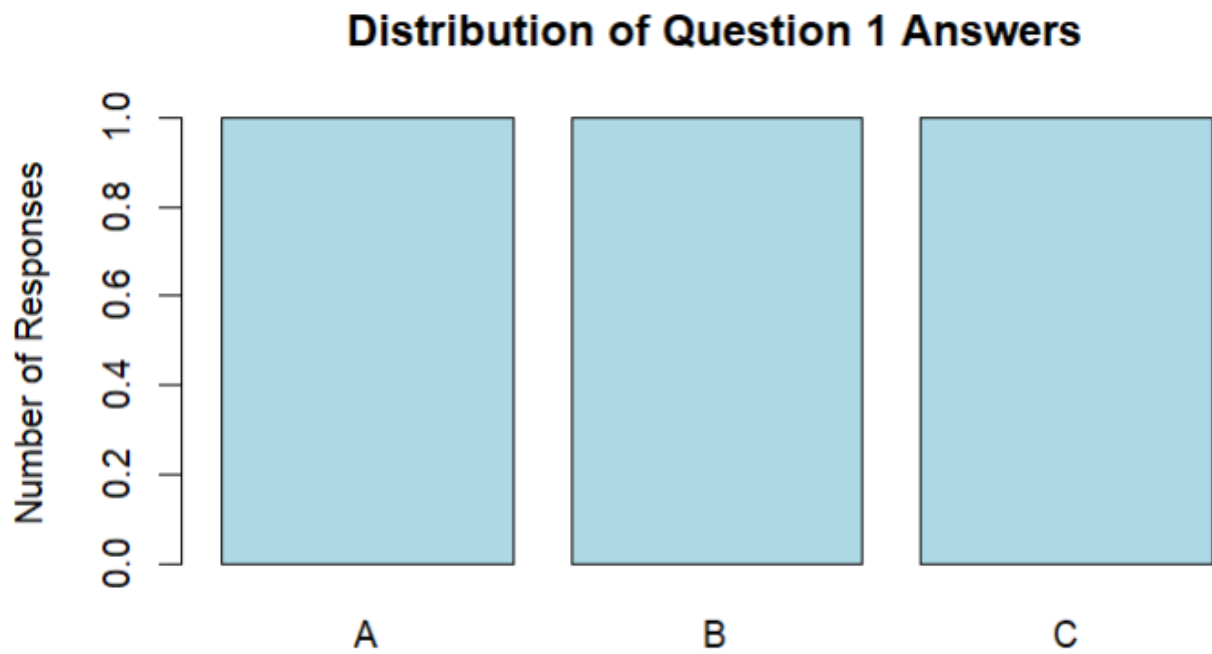
Question 3 - Table

```
print(survey)
```

Question 1

Create a stacked bar chart to display the distribution of answers for Question 1.

Output 1

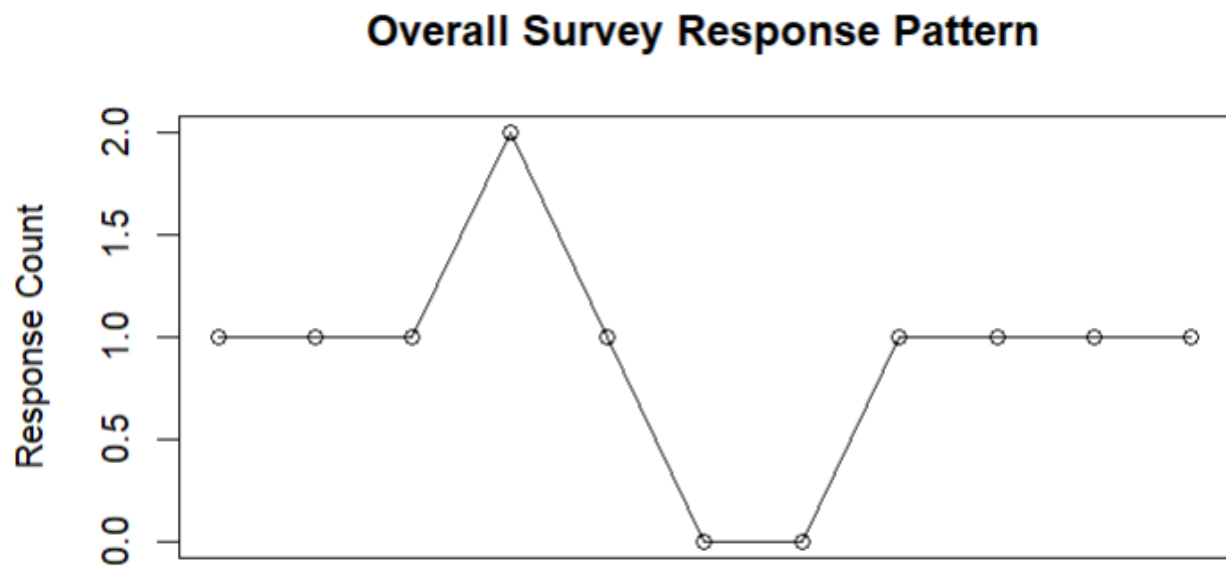


The chart displays one response each for A, B, and C.

Question 2

Generate a radar-style chart to visualize the overall response pattern.

Output 2



The radar-style visualization displays the frequency of responses across the survey questions.

Question 3

Display the survey response data in a table.

Output 3

```
> # Question 3 - Table
> print(survey)
  Respondent Question1 Question2
1           1         A         B
2           2         B         A
3           3         C         A
  Question3
1          C
2          D
3          B
```